# SIGAN: Spectral Index Generative Adversarial Network for Data Augmentation in Multispectral Remote Sensing Images

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***Abstract*— Generative models are typically employed to approximate the distribution of deep features. Recently, these state-of-the-art methods have been applied to estimate image transformations by an unsupervised learning approach. In this letter, a novel spectral index generative adversarial network (SIGAN) is proposed for the generation of multispectral (MS) remote sensing images. This network is defined to effectively perform data augmentation starting from a limited number of training samples in the MS remote sensing domain for training deep learning models. The SIGAN model is able to capture class-specific properties in data augmentation, by incorporating the task-specific normalized spectral indices to model class-by-class properties of MS images. Experimental results obtained on a Sentinel 2 dataset show that the proposed model provides better performance than other generative adversarial networks (GANs) in MS data generation.**

***Index Terms*— Data augmentation, labeled sample, multispectral (MS) images, remote sensing.**

## I. Introduction

RESEARCHERS have used deep learning methods for many tasks in the wider field of computer vision, such as image recognition, image classification, semantic segmentation, and object detection. Within these tasks, deep neural networks are generally used to learn abstract information with several levels of representation from lower to higher levels. In this context, deep learning is being increasingly used in the analysis of satellite remote sensing data. A foundation for estimating generative tasks through generative adversarial networks (GANs) has been introduced in [1]. It consists of a generative model and a discriminative model, which are trained simultaneously. A discriminator is trained to categorize between real and fake samples and the generator to fool the discriminator by generating real-looking samples. GANs exploit the min-max operator in which the discriminator aims to maximize the objective function, while the generator aims to minimize it. The min-max operator minimizes the divergence between the data and the distribution of deep features [1]. A conditional version of GANs can control the method used for data generation, incorporating additional information, such as class labels, to monitor the artificial data generation process [2]. In conditional GANs (CGAN), conditional information can be given as input to both the generator and the discriminator.

In [3], a deep convolutional GAN (DCGAN) is presented to learn an unsupervised representation of images for generative modeling. The DCGAN uses convolutional neural networks (CNNs) for generator and discriminator and was successful in learning hierarchical representations. A pixel-to-pixel mapping is presented in [4], which utilizes the CGAN as a tool for the image-to-image translation by conditioning the input images to generate synthetic output images. This image-to-image translation approach has also the capability to learn the loss function while training the model. Several experiments are presented in which image-to-image translation was successful in a range of tasks, including colorization, image reconstruction, and image synthesis [4].

The learning of deep models for satellite remote sensing image analysis requires a large number of training samples to estimate the very large number of parameters included in deep architectures [5]. However, often a reduced amount of reliable labeled data is available, thus creating a problem for the estimation of deep model parameters with good prediction capability. At present, there are no reliable pretrained deep learning architectures that can be used to solve the challenges of training with a relatively small number of samples by incorporating spectral and spatial properties of the remote sensing images [6]. Researchers have worked on the augmentation of data through padding, cropping, zooming, and flipping to provide sufficient data for the training of deep neural networks. However, these basic augmentation techniques do not provide diverse data for effective training of deep learning models. Most of the research works regarding data augmentation with GAN have been developed for general datasets. Only a few recent studies considered the use of GANs for data augmentation in remote sensing [7]–[10]. For example, in [11], a deep generative model is proposed for synthetic aperture radar (SAR) image generation. This is an extension of the Wasserstein autoencoder that resulted successful in improving reconstruction loss and object recognition accuracy. In [12], a two-stage GAN was proposed to generate synthetic radar image sequences. Some studies have analyzed generative models for image-to-image translation for paired and unpaired images and successfully synthesized images from label maps and edge maps [4], [13]. In this context, it is crucial to improve the capability to generate realistic samples for the learning of deep architectures.

To address the aforementioned issue, we propose a technique inspired by the concept of image-to-image translation approach introduced in [4]. In particular, we present a novel spectral index GAN (SIGAN) for the reliable generation of training data. The proposed SIGAN generates synthetic

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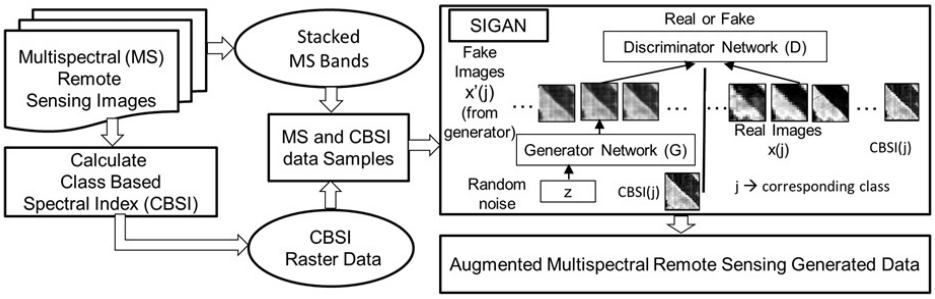



Fig. 1. Methodology adopted for MS data generation using the proposed SIGAN.

multispectral (MS) images by incorporating the normalized class-based spectral index (CBSI) information corresponding to each MS image in the generator network. Results obtained on MS Sentinel-2 data point out the efficiencies of the SIGAN technique with respect to other GANs. This letter is divided into four sections. Section II describes the proposed SIGAN. Experimental results are presented and analyzed in Section III. Finally, Section IV concludes this letter.

## II. Proposed SIGAN

The proposed method is specifically defined for dealing with MS remote sensing data. We generate synthetic MS images using SIGAN by exploiting a normalized spectral index to better capture the properties of each class. The method incorporates class-specific information through the spectral index to generate synthetic MS samples that can be used for data augmentation to train deep learning models for applications such as image classification. Fig. 1 shows the methodology adopted for MS data generation using the proposed SIGAN. Let us consider stacked MS remote sensing images and a set of related spectral indices corresponding to each image defined to be given as input to the proposed SIGAN. CBSI is used to calculate the spectral index between the bands that characterize the behavior of the class in terms of high reflectance and low reflectance, that is

$$\text{CBSI} = \frac{b_h - b_l}{b_h + b_l} \tag{1}$$

where $b_h$ and $b_l$ are the high and low reflectance spectral bands, respectively, of the considered MS image. Let $x(j) = \{x_1, x_2, x_3, x_4, \ldots, x_p\}$ be the set of real MS images of the $j$th class and CBSI $(j) = \{\text{CBSI}_1, \text{CBSI}_2, \text{CBSI}_3, \text{CBSI}_4, \ldots, \text{CBSI}_p\}$ be the set of spectral index values of each image. Different spectral indices can be defined for the different classes to emphasize their class properties. In this research, normalized difference vegetation index (NDVI), normalized difference water index (NDWI), and normalized difference built index (NDBI) are calculated for classes, such as vegetation (grassland, conifers), water, and artificial land, respectively. Other specific indices can be used for other classes (see Table I for some examples).

Inspired by the deep convolution approach, the proposed SIGAN consists of a deep convolution generator and a discriminator. The SIGAN generator and discriminator consist of convolutional layers followed by batch normalization and Leaky rectified linear unit (LRELU) activation functions. After the convolution of the input with the kernel, features generated in the generator and discriminator are passed through nonlinear Leaky RELU activation functions. Between the convolution and activation operations, batch normalization is used for faster convergence and to avoid overfitting and saturation. For the training and optimization process of SIGAN generator and discriminator, the Adam optimizer is used. Fig. 2 shows

TABLE I

Examples of Normalized Spectral Indices That Can Be Used for Specific Classes. (NDVI—Normalized Difference Vegetation Index; NDWI—Normalized Difference Water-Up Index; NDBI—Normalized Difference Built-up Index; BSI—Bare Soil Index; and NBR—Normalized Burned Ratio)

| Spectral Index | Definition | Class |
|---|---|---|
| NDVI | $\frac{b_{NIR} - b_{RED}}{b_{NIR} + b_{RED}}$ | Vegetation |
| NDWI | $\frac{b_{GREEN} - b_{SWIR}}{b_{GREEN} + b_{SWIR}}$ | Water bodies |
| NDBI | $\frac{b_{SWIR} - b_{NIR}}{b_{SWIR} + b_{NIR}}$ | Artificial land |
| BSI | $\frac{(b_{RED} + b_{SWIR}) - (b_{BLUE} + b_{NIR})}{(b_{RED} + b_{SWIR}) + (b_{BLUE} + b_{NIR})}$ | Bareland |
| NBR | $\frac{b_{NIR} - b_{SWIR}}{b_{NIR} + b_{SWIR}}$ | Burnt Area |

**Algorithm 1** Training of the Proposed SIGAN Algorithm

**Generator Input**: Vector of random array and class based spectral index information
**Discriminator Input**: Real multi-spectral images with class based spectral index information and multi-spectral generated data
**Generator Model**: G, **Discriminator Model**: D, **Batch of Real Data**: x(j), **Batch of Class Based Spectral Index**: CBSI(j), **Batch of Generated data**: x'(j);
**for** epoch = 1 to N: **do**
  **for** x and CBSI image pairs in range (batch): **do**
    → Assign random batch of real dataset-spectral index pair (x|CBSI) to the disciminator;
    → Compute D(x|CBSI) for the batch;
    → Back-propagate the discriminator loss;
    → Assign batch of random noise vectors and class based spectral index (z|CBSI) to the generator;
    → Generate batch of fake samples: G(z|CBSI)= x'|CBSI;
    → Compute D(x, x'| CBSI) for the batch;
    → Back-propagate the discriminator and generator loss

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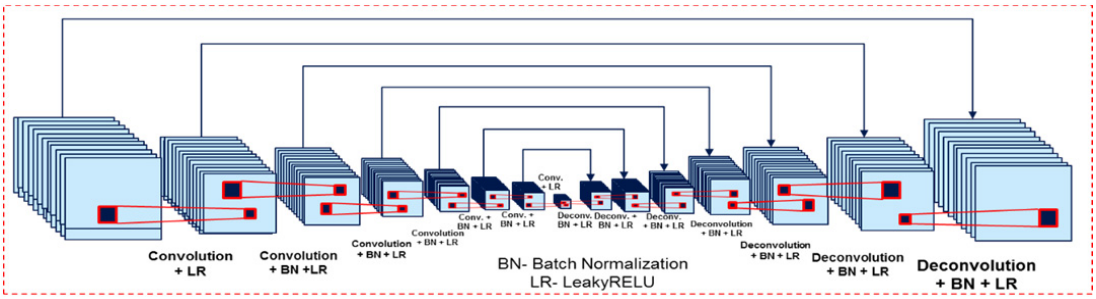



Fig. 2. Architecture of the SIGAN Generator.

the architecture of the SIGAN Generator. Each block in the encoder of the generator includes Convolution → Batch Normalization → LeakyRELU, while the decoder includes Transposed Convolution → Batch Normalization → LeakyRELU. In the first three blocks of decoder, dropout is used. The generator loss is calculated between the generated image and the real image. The discriminator loss function takes generated images and real images to calculate the loss. The cost function of the proposed SIGAN is defined as follows:

$$\mathcal{L}_{\text{SIGAN}}(G, D) = \min_{G}\{\max_{D}\{\mathbb{E}_{x,\text{CBSI}}[\log(D(x|\text{CBSI}))] + \mathbb{E}_{x,z}[\log(1 - D(x, G(z|\text{CBSI}))]\}\} \quad (2)$$

where $D(x|\text{CBSI})$ and $G(z|\text{CBSI})$ are the discriminative and generative image samples, respectively, CBSI is the CBSI selected for the specific class considered, $x$ represents the input real MS image, and $z$ is the noise used for generating synthetic training samples. The resulting SIGAN is used for data augmentation. A prior random noise ($z$) and a CBSI information are given as input to the generator to produce the output ($x'$). The real data ($x$), generated data ($x'$), and CBSI information are feed to the discriminator model to classify the real and generated data. Algorithm 1 describes the training of the proposed SIGAN algorithm.

The goal of the generator of the proposed SIGAN is to generate realistic fake MS images by incorporating spectral index information. The goal of the discriminator of the SIGAN is to distinguish fake generated images and spectral index pairs from the generator, and real images and spectral index pairs from the training set.

Note that the spectral index exploits two spectral channels out of all the available channels to better focus on the specific properties of the classes. This improves the quality of produced images without affecting the diversity of the generated samples and the generalization ability of SIGAN, which remains smaller than that of other GANs.

## III. Experimental Results

### A. Dataset Description and Model Setup

Sentinel-2 MS images have been considered for the experiment. The training samples were collected from areas all over Italy, which includes various classes including grassland, conifers, water, artificial land, bareland, broadleaves, cropland, and shrubland [14]. We sampled the data according to the Land Use and Cover Area frame Survey (LUCAS) map, which provides a single-point label. The LUCAS survey is usually carried out every three years since 2006 over all 27 EU countries, which analyzes different land cover types over 270 000 points. For preliminary experiments, we considered

TABLE II
Computation Time (Per Epoch) of Different GANs

| Method | Computation Time (per epoch) | |
|---|---|---|
| | Water | Artificial Land |
| GAN | 797.28 s | 869.28 s |
| WGAN | 427.59 s | 471.08 s |
| SIGAN | 401.14 s | 434 s |

12 spectral bands of Sentinel-2 with spatial resolution variable from 10 to 60 m and a radiometric resolution of 12 bit. The number of training samples of water, grassland, conifers, and artificial land is 70, 587, 1005, and 346, respectively.

Let us now analyze the model setup. The generator has 16 layers, including 15 convolution and deconvolution layers, and an output layer. For the training of the generator, the Adam optimizer was used with a learning rate of $2 \times 10^{-4}$. Stride $= 2 \times 2$, dropout rate $= 0.5$, and kernel size $= 7 \times 7$ were fixed. We have taken LeakyRELU for fast computation and fast convergence. The mean absolute error and mean square error were used for calculating the generator loss after each epoch. The discriminator has five layers, including four convolution layers and an output layer. In the case of the discriminator, the Adam optimizer was used with a learning rate of $2 \times 10^{-4}$. Stride $= 2 \times 2$ and kernel size $= 3 \times 3$ were fixed. LeakyRELU was used for fast computation and fast convergence. Binary cross entropy has been used to analyze discriminator accuracy and discriminator loss. We used the binary cross-entropy approach for binary classification (real versus fake). During the training of the proposed SIGAN, the batch size was set between 10 and 50 and the number of epochs was between 300 and 400 for classes, such as water, grassland, conifers, and artificial land classes. Fig. 3 shows the behavior of the loss function of the SIGAN versus the number of epochs for different values of the learning rate, the dropout, and the kernel size.

### B. Results

Qualitative and quantitative results have been analyzed during the training process of the proposed SIGAN. Fig. 4 shows the generator loss versus the number of epochs for grassland, conifers, water, and artificial land classes. SIGAN obtained more accurate results than the standard GAN during the training process for all the cases. Table II shows the computation time (per epoch) for water and artificial land class. Fig. 5 shows the synthetic MS data generated for the grassland training samples. The results are shown after 50 and 300 epochs. Columns 1–12 depict the spectral bands

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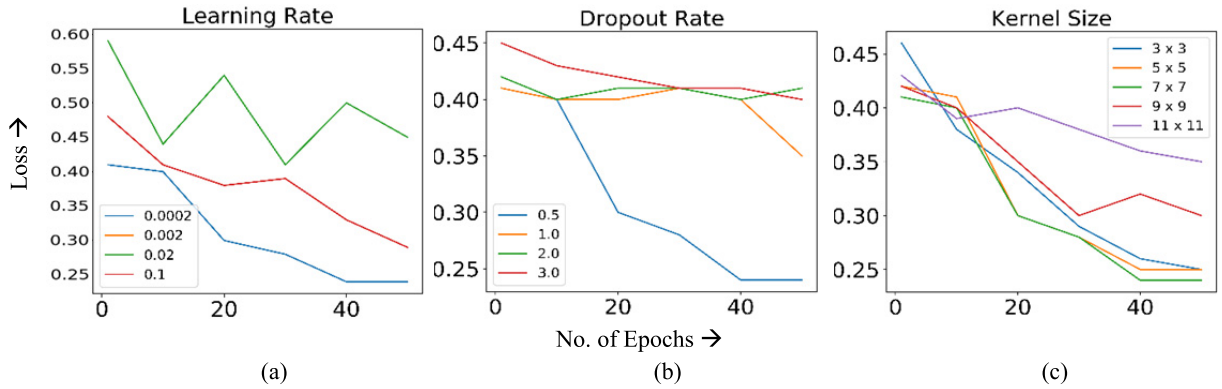



Fig. 3. Behavior of the loss function of the SIGAN for different values of (a) learning rate, (b) dropout rate, and (c) kernel size.

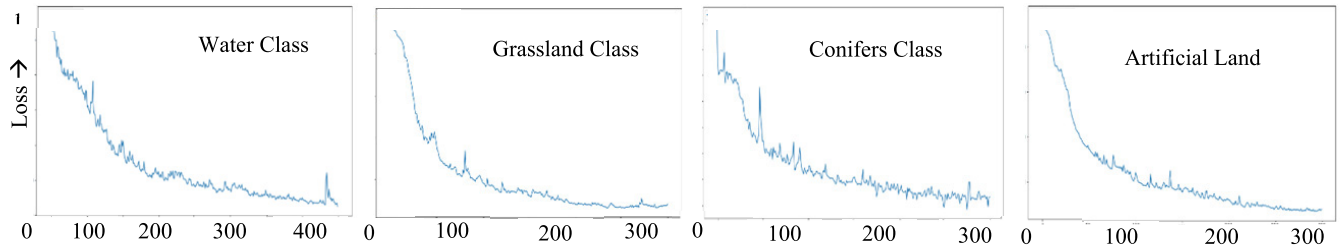



Fig. 4. Generator loss versus the number of epochs for grassland, conifers, artificial land, and water classes.

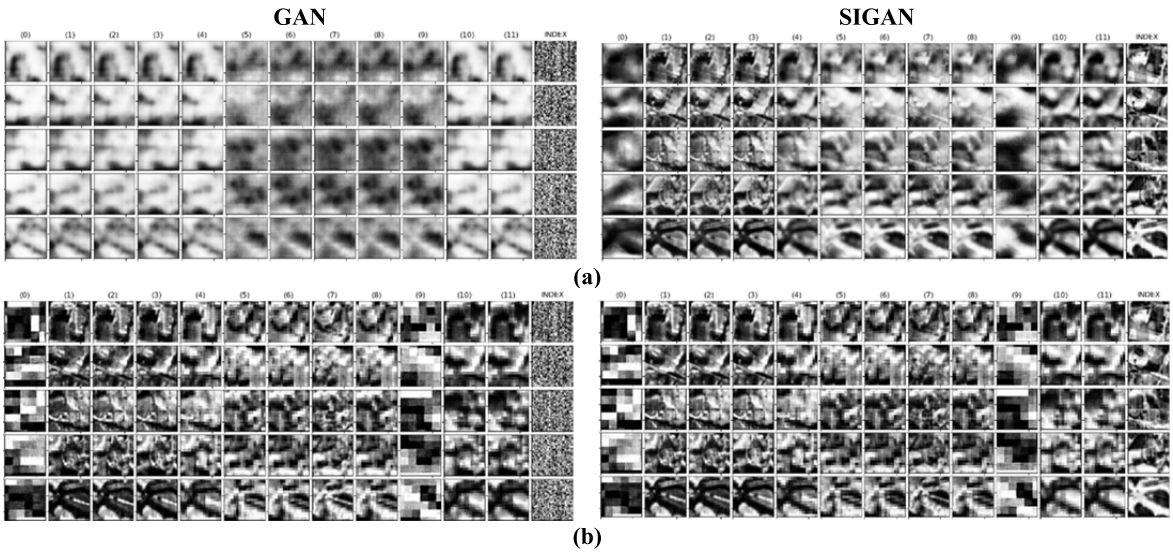



Fig. 5. Synthetic MS data generated using the standard GAN and the proposed SIGAN for the grassland class after (a) 50 epochs and (b) 300 epochs. Different columns represent different spectral bands of Sentinel 2. The last column shows the noise or CBSI.

TABLE III

EXPERIMENTAL RESULTS PROVIDED BY A STANDARD GAN, THE WGAN, AND THE PROPOSED SIGAN

| Performance Metrics | Class | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Water | | | Grassland | | | Conifers | | | Artificial Land | | |
| | GAN | WGAN | SIGAN | GAN | WGAN | SIGAN | GAN | WGAN | SIGAN | GAN | WGAN | SIGAN |
| Discriminator/Critic Loss | 0.22 | 0.29 | 0.31 | 0.22 | 0.25 | 0.44 | 0.20 | 0.31 | 0.41 | 0.25 | 0.32 | 0.34 |
| Generator Loss | 0.16 | 0.14 | 0.11 | 0.14 | 0.15 | 0.12 | 0.13 | 0.13 | 0.11 | 0.13 | 0.15 | 0.13 |

from 1 to 12, while column 13 represents the noise/CBSI. From the figure, one can see that the proposed SIGAN method is computationally efficient as it takes few epochs to generate a texture similar to the corresponding real image. Table III reports the comparison of the proposed SIGAN with the Wasserstein GAN (WGAN) and a standard GAN in terms of discriminator/critic loss and generator loss. Note that as a substitute for the discriminator to categorize the probability of generated images as real or fake, the WGAN exploits a critic model that scores the realness or fakeness of a given

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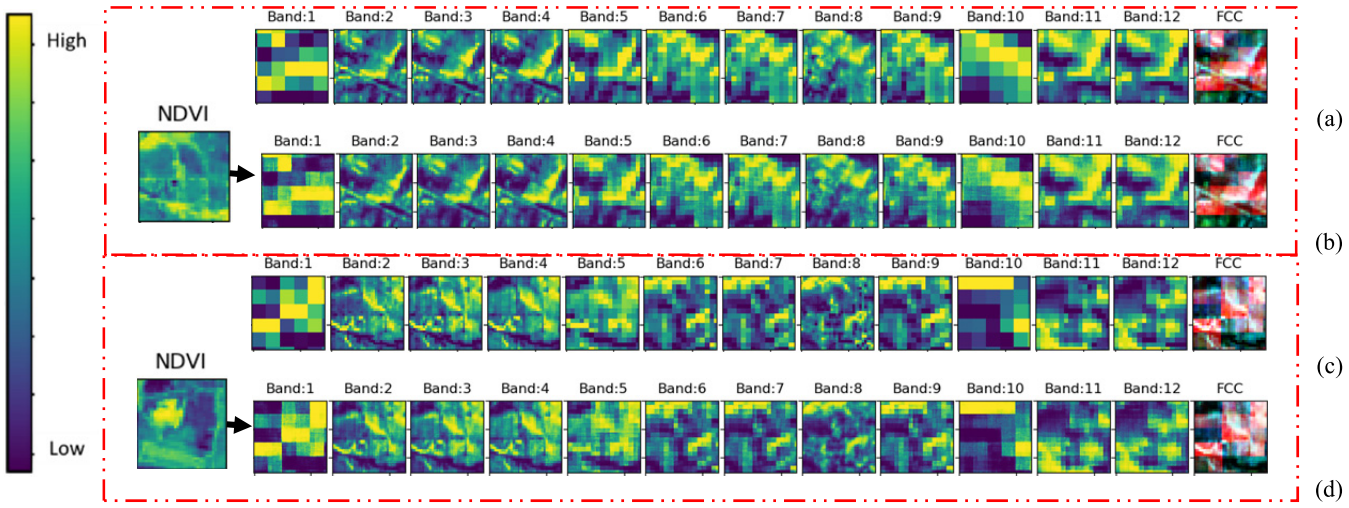



Fig. 6. (a) and (c) Real MS data. (b) and (d) Predicted artificial MS from NDVI input using the SIGAN model.

image. Results confirm that the loss function implemented in the proposed SIGAN is more reliable compared to the Wasserstein loss for generating real-looking MS samples. The discriminator accuracy obtained by the SIGAN after training is 0.68, 0.63, 0.62, and 0.60 for Water, Grassland, Conifers, and Artificial Land classes, respectively. The discriminator accuracy in the case of the standard GAN is 0.80, 0.76, 0.81, and 0.79 for Water, Grassland, Conifers, and Artificial Land classes, respectively. This confirms the effectiveness of the SIGAN.

Fig. 6(a) and (c) shows the real MS images and Fig. 6(b) and (d) shows its artificial MS data from NDVI input using the SIGAN model. Columns 1–12 depict the spectral bands from 1 to 12, while column 13 represents the false-color composite near-infrared response (NIR, red, and green) of the MS image. It is possible to see that all the spectral bands of Sentinel 2 with spatial resolution between 10 and 60 m are correctly generated. This points out the high impact of using the spectral index approach during the training of the SIGAN model. Accurate results have also been observed in classes with few samples, such as water and artificial land.

## IV. Conclusion

In this letter, we have presented an SIGAN for synthetic MS remote sensing data generation. The main novelty of the proposed SIGAN consists in capturing class-specific properties of MS images by using dedicated normalized spectral indices, which are different for each class in order to properly emphasize the spectral properties of classes. Results confirm that SIGAN is significantly more accurate than other GANs when applied to MS data. The generated data are very useful to define augmented MS training data. Data augmentation using SIGAN has been demonstrated on a Sentinel-2 MS dataset. We have compared the proposed SIGAN with standard GAN and evaluated the related performances. The SIGAN obtained a sharp improvement in both generator loss and discriminator loss. Moreover, it converged faster compared to standard GAN. As future work, we plan to use the high-quality generated data as training samples in supervised classification, object detection, and change detection.

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